Two-Phase Deep Learning for Deepfake and AI-Generated Video Detection

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Introduction

This project analyzes video authenticity by classifying content as real, deepfake, or AI-generated. With rapid advances in generative AI, fake videos have become more realistic and harder to detect. The proposed system uses a unified approach to identify both manipulated and AI-generated videos effectively.

Motivation

The rapid rise of AI-generated videos has increased risks such as misinformation, fraud, and loss of trust in digital content. This project aims to develop a reliable system for verifying video authenticity in real-world applications like social media, journalism, and security.

Scope of the Project

This project focuses on analyzing facial videos using a two-phase classification includes camera vs non-camera and real vs deepfake . It includes key steps such as frame extraction, face detection, key preprocessing steps , feature extraction, and classification using deep learning models. The system categorizes videos into three classes: Pristine , deepfake (manipulated), and AI-generated content(non-camera). Phase 1 aims at classifying video level and phase 2 is designed to be efficient and focused on facial regions , where most manipulations occur .

Methodology

The proposed system follows a two-phase hierarchical pipeline for video authenticity detection, as illustrated in the figure 1.Initially, the input video may belong to three categories: real, AI-generated, or manipulated. In Phase 1, the system extracts frames and analyzes them using residual features and frequency-based techniques like FFT to distinguish between AI-generated (non-camera) and camera-captured videos , by analyzing any camera footprints are there or not.

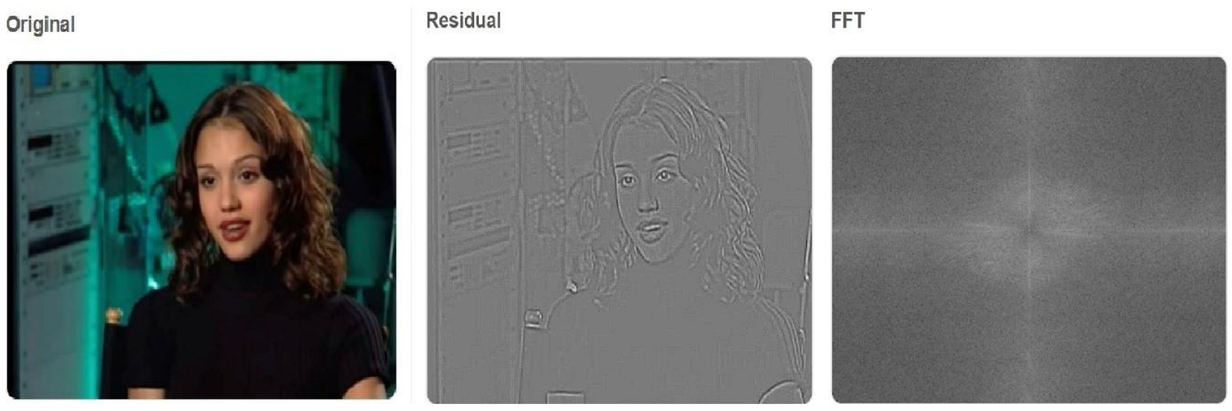

Figure 1: *Comparison of original video frame, residual representation and FFT-based frequency spectrum .*

If the video is identified as non-camera (AI-generated), it is directly classified as synthetic content. However, if it is a camera-captured video, it proceeds to Phase 2 for further analysis. In this phase, facial regions are detected using MTCNN, and important features are extracted from the face. These features are then processed using CNN Ensemble of XceptionNet and EffeciantNet baseline to classify the video as either real (authentic) or fake (deepfake manipulated).

This two-stage approach given in figure 2 enables a more refined and accurate classification compared to traditional methods by first separating AI-generated content and then detecting deepfake manipulations within real-world videos.

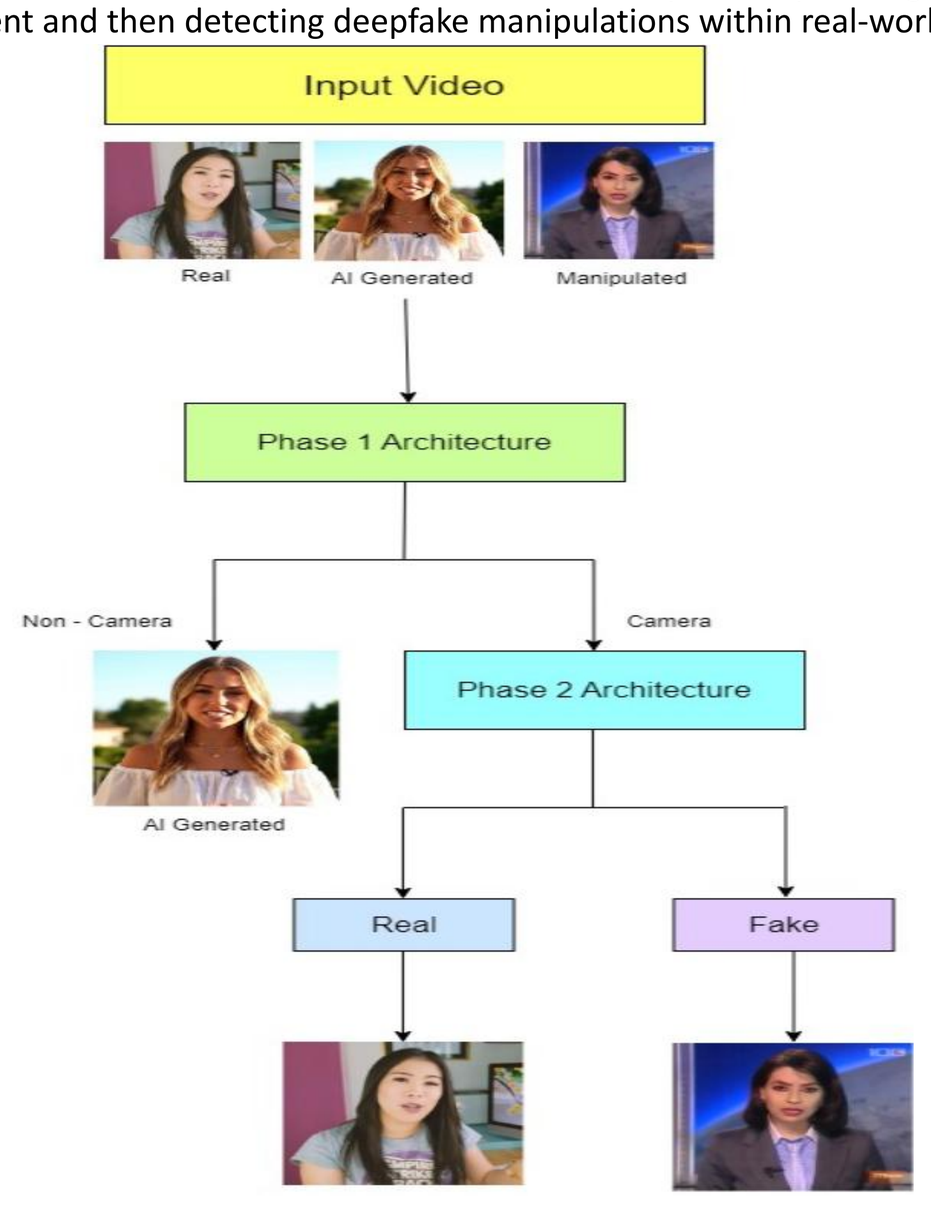

Figure 2: *Two-phase architecture for video authenticity detection*

Results

The proposed system classifies videos into real, deepfake, and AI-generated categories by combining spatial and frequency-based features, resulting in improved detection accuracy. The two-phase approach enhances precision by first separating AI-generated content and then detecting manipulations, reducing misclassification.

By leveraging FFT and residual features, the system effectively identifies subtle artifacts and maintains consistent performance across different datasets and conditions such as compression, lighting, and video quality. It also provides predictions with confidence scores, making it reliable and suitable for real-world applications like digital forensics, social media monitoring, and security systems.

Class	Precision	Recall	F1-Score
AI	1.00	0.99	0.99
Camera	0.99	1.00	1.00

Performance Metrics of Phase 1 Model for Camera vs AI Classification (Precision, Recall, and F1-Score) and overall accuracy was 99.5% .

Model	CelebDF	DFDC	FaceForensics++
Xception Baseline	96.25	89.23	100.00
EfficientNet-B2	93.13	87.69	100.00
Ensemble	96.88	92.31	100.00

Overall and Dataset-wise Test Accuracy Comparison .The Ensemble model achieves the highest overall accuracy of 96.67%, followed by Xception Baseline at 95.56% and EfficientNet-B2 at 94.00%.

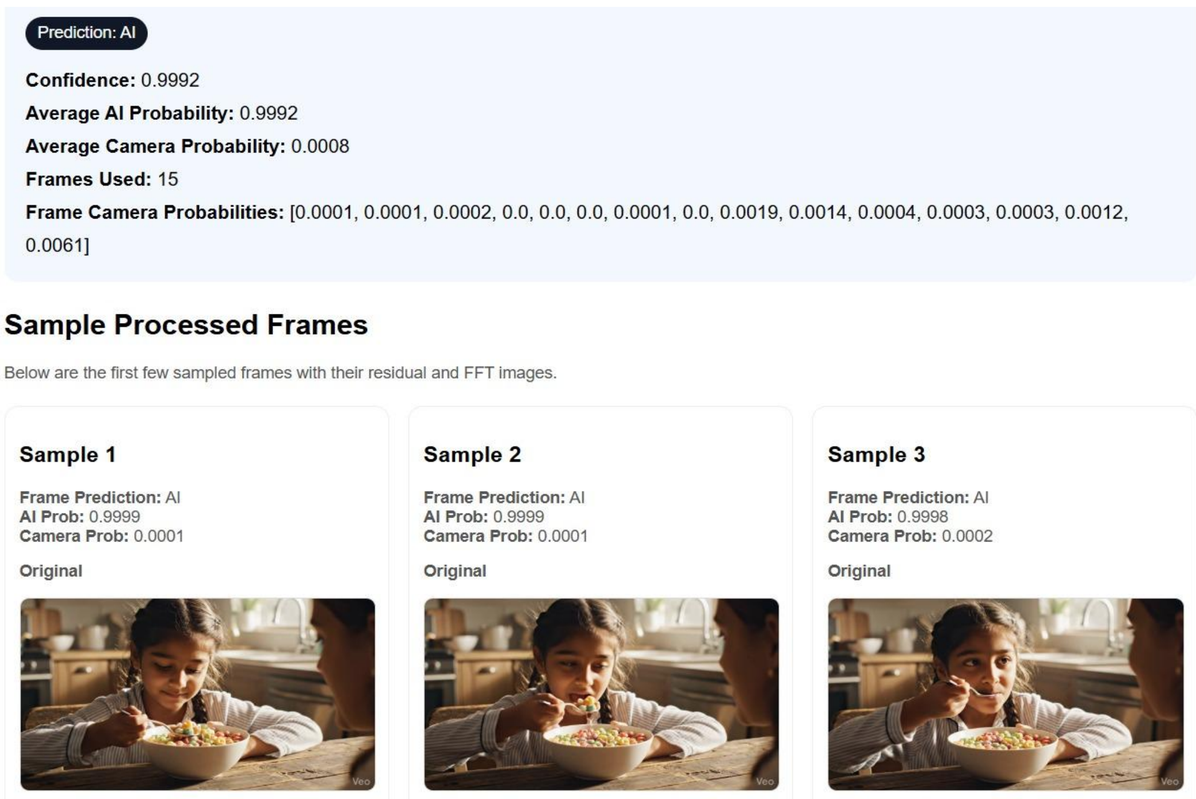

Figure: *Model output showing video classified as camera-captured with high confidence, along with frame-level predictions and sample processed frames used for analysis*

Conclusion

This project presents a two-phase framework to classify videos as real, deepfake, or AI-generated. By combining spatial and frequency-based features, the system improves detection accuracy and captures subtle inconsistencies.

The hierarchical approach enhances performance by separating AI-generated content and then detecting deepfakes in camera-captured videos. Overall, the system provides a reliable and practical solution for applications such as media verification, digital forensics, and security.

References

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